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Table S1. Detailed Queries

Database	Queries
Embase	('mortality'/exp OR 'death'/exp OR 'cardiovascular disease'/exp OR 'coronary heart disease'/exp OR 'acute coronary syndrome'/exp OR 'myocardial infarction'/exp OR 'cerebrovascular event') AND 'cholesterol efflux capacity':ab AND ([article]/lim OR [article in press]/lim OR [conference abstract]/lim OR [conference paper]/lim) AND [english]/lim AND ([embase]/lim OR [medline]/lim OR [embase classic]/lim OR [pubmed-not-medline]/lim)
PubMed	cholesterol efflux capacity[tw] AND (mortality[tw] OR death[tw] OR cardiovascular disease[tw] OR coronary heart disease[tw] OR acute coronary syndrome[tw] OR myocardial infarction[tw] OR stroke[tw] OR cerebrovascular event[tw]) AND Journal Article[pt]
Web of Science	(((TS =(cholesterol efflux capacity)) AND ((TS =(mortality OR death OR cardiovascular disease OR coronary heart disease OR acute coronary syndrome OR myocardial infarction OR stroke OR cerebrovascular event))))) AND LANGUAGE: (English) AND DOCUMENT TYPES: (Article OR Abstract of Published Item)

Table S2. Quality Assessment Criteria for Case-Control Studies

Domain	Item	Criterion	Rating
Selection	Is the Case Definition Adequate? (0 to 2)	Requires some independent validation (e.g. >1 person/record/time/process to extract information, or reference to primary record source such as x-rays or medical/hospital records).	2
		Record linkage (e.g. ICD codes in database) or self-report with no reference to primary record.	1
		No description.	0
	Representativeness of the Cases (0 to 1)	All eligible cases with outcome of interest over a defined period of time, all cases in a defined catchment area, all cases in a defined hospital or clinic, group of hospitals, health maintenance organization, or an appropriate sample of those cases (e.g. random sample).	1
		Not satisfying abovementioned requirements with potential for selection biases, or not stated.	0
	Selection of Controls (0 to 1)	Community controls (i.e. same community as cases and would be cases if had outcome).	1
		Hospital controls, within same community as cases (i.e. not another city) but derived from a hospitalized population.	1
		No description.	0
	Definition of Controls (0 to 1)	If cases are first occurrence of outcome, then it must explicitly state that controls have no history of this outcome. If cases have new (not necessarily first) occurrence of outcome, then controls with previous occurrences of outcome of interest should not be excluded.	1
		No mention of history of outcome.	0
Comparability	Matched or Adjusted by Key Factors (0 to 1)	Matched or adjusted by key factors including age, sex, and cardiovascular risk factors.	1
	Matched or Adjusted by Additional Factors (0 to 1)	Matched or adjusted by additional factors including total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, or triglycerides.	1
Exposure	Ascertainment of Exposure (0 to 1)	Secure records (e.g. medical records/research records)	1
		Self-reported records only, or not stated.	0
	Non-Response Rate (0 to 1)	Non-response rate is described.	1
		No description.	0

Table S3. Quality Assessment Form for Case-Control Studies

Study	Selection				Comparability		Outcome		Total Score
	Is the Case Definition Adequate? (0 to 2)	Representativeness of the Cases (0 to 1)	Selection of Controls (0 to 1)	Definition of Controls (0 to 1)	Matched or Adjusted by Key Factors (0 to 1)	Matched or Adjusted by Additional Factors (0 to 1)	Ascertainment of Exposure (0 to 1)	Non-Response Rate (0 to 1)	
Ebtehaj 2019	1	1	1	1	1	1	1	1	8
Cahill 2019	2	0	1	1	1	1	1	1	8
Tejera-Segura 2017	1	1	0	1	1	1	1	0	6
Khera 2017	1	1	1	1	1	1	1	1	8
Gall 2016	1	1	1	1	1	1	1	1	8
Ishikawa 2015	2	1	1	1	1	1	1	0	8
Saleheen 2015	1	1	1	1	1	1	1	0	7
Li 2013	1	1	1	1	1	1	1	0	7

Table S4. Quality Assessment Criteria for Cohort Studies

Domain	Item	Criterion	Rating
Selection	Representativeness of the Exposed Cohort (0 to 1)	Exposed individuals are representative in the community.	1
		No description.	0
	Selection of the Non-Exposed Cohort (0 to 1)	Drawn from the same community as the exposed cohort.	1
		No description.	0
	Ascertainment of Exposure (0 to 1)	Secure records (e.g. medical records/research records).	1
		No description.	0
	Demonstration That Outcome of Interest Was Not Present at Start of Study (0 to 1)	Yes	1
		No description.	0
Comparability	Matched or Adjusted by Key Factors (0 to 1)	Matched or adjusted by key factors, including age, sex, and cardiovascular risk factors.	1
	Matched or Adjusted by Additional Factors (0 to 1)	Matched or adjusted by additional factors including total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, or triglycerides.	1
Outcome	Assessment of Outcome (0 to 1)	Independent or blind assessment, confirmation of the outcome by reference to secure records (e.g. medical records/research records), or record linkage (e.g. identified through ICD codes on database records).	1
		Self-reported records only, or not stated.	0
	Was Follow-Up Long Enough for Outcomes to Occur (0 to 1)	Yes (at least one year).	1
		Less than one year or no description.	0
	Adequacy of Follow Up of Cohorts (0 to 1)	At least 80% of subjects complete the follow-up, and subjects lost to follow-up are unlikely to introduce bias.	1
		Follow-up rate < 80% and no description of subjects lost to follow-up.	0

Table S5. Quality Assessment Form for Cohort Studies

Study	Selection				Comparability		Outcome			Total Score
	Representativeness of the Exposed Cohort (0 to 1)	Selection of the Non-Exposed Cohort (0 to 1)	Ascertainment of Exposure (0 to 1)	Demonstration That Outcome of Interest Was Not Present at Start of Study (0 to 1)	Matched or Adjusted by Key Factors (0 to 1)	Matched or Adjusted by Additional Factors (0 to 1)	Assessment of Outcome (0 to 1)	Was Follow-Up Long Enough for Outcomes to Occur (0 to 1)	Adequacy of Follow Up of Cohorts (0 to 1)	
Guerin 2018	0	1	1	1	1	1	1	1	0	7
Chindhy 2018	1	1	1	1	1	1	1	1	0	8
Bauer 2017	1	1	1	1	1	1	0	1	0	7
Kopecky 2016	1	1	1	0	1	1	1	1	0	7
Javaheri 2016	1	1	1	0	1	1	1	1	0	7
Mody 2016	1	1	1	1	1	1	1	1	1	9
Liu 2016	1	1	1	1	1	1	1	1	1	9
Zhang 2016	0	1	1	1	1	1	1	1	1	8
Ogura 2016	0	1	1	0	1	1	1	0	0	5
Annema 2016	0	1	1	1	1	1	1	1	1	8
Rohatgi 2014	1	1	1	1	1	1	1	1	1	9
Khera 2011	1	1	1	1	1	1	1	0	0	7

Table S6. Summary of Egger's Test for Small-Study Effect and Trim-and-Fill Method for Publication Bias

	Egger's Test for Small-Study Effects				Effect of Observed Studies				Effect of Observed plus Imputed Studies			
Endpoint	**β1**	**SE**	**z**	**P-value**	**Number of Studies**	**Point Estimate**	**95% LCL**	**95% UCL**	**Number of Studies**	**Point Estimate**	**95% LCL**	**95% UCL**
Adverse Cardiovascular Event: High CEC vs. Low CEC (RR)	-3.84	0.852	-4.51	<0.0001	14	0.631	0.522	0.762	18	0.790	0.646	0.966
Adverse Cardiovascular Event: Per SD Increment of CEC (adjusted HR)	-2.79	0.837	-3.33	0.0009	10	0.758	0.628	0.914	12	0.807	0.664	0.980
Atherosclerotic Cardiovascular Disease: High CEC vs. Low CEC (RR)	-3.58	0.870	-4.11	<0.0001	12	0.662	0.550	0.797	15	0.775	0.638	0.941

Abbreviations: CEC, cholesterol efflux capacity; HR, hazard ratio; LCL, lower confidence limit; RR, risk ratio; SD, standard deviation; SE, standard error; UCL, upper confidence limit.

Figure S1. PRISMA Flow Diagram

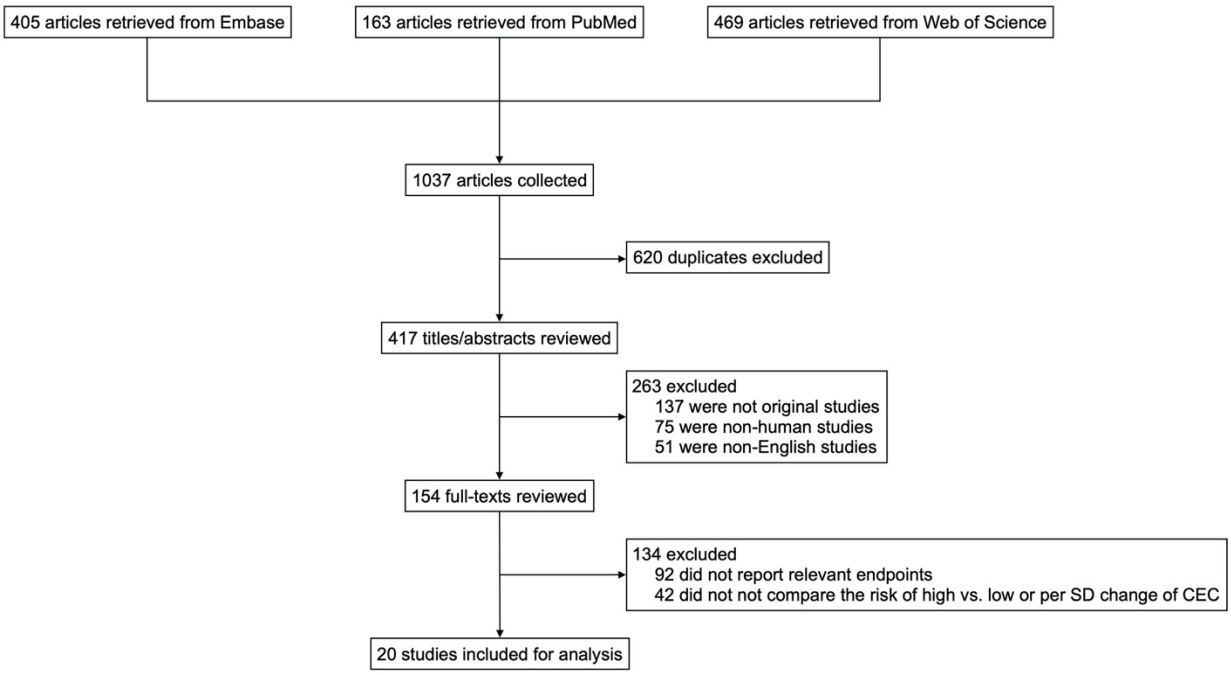

Figure S2. Atherosclerotic Cardiovascular Disease: High CEC vs. Low CEC (RR) (12 studies)

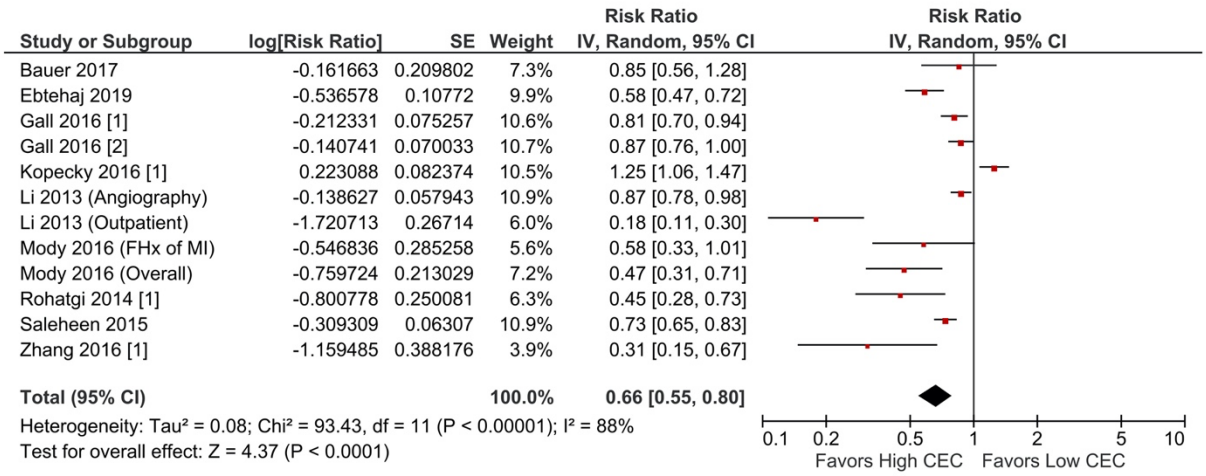

Figure S3. Atherosclerotic Cardiovascular Disease: High CEC vs. Low CEC (Adjusted RR) (7 studies)

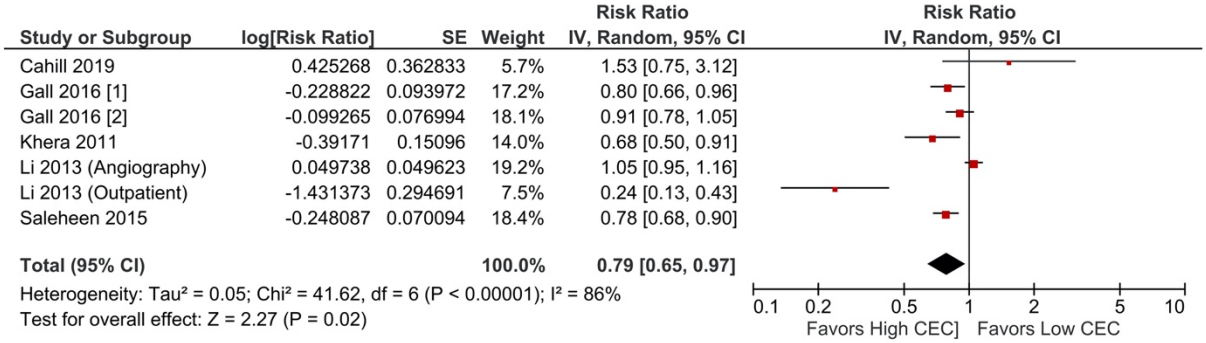

Figure S4. Atherosclerotic Cardiovascular Disease: Per SD Increment of CEC (HR) (2 studies)

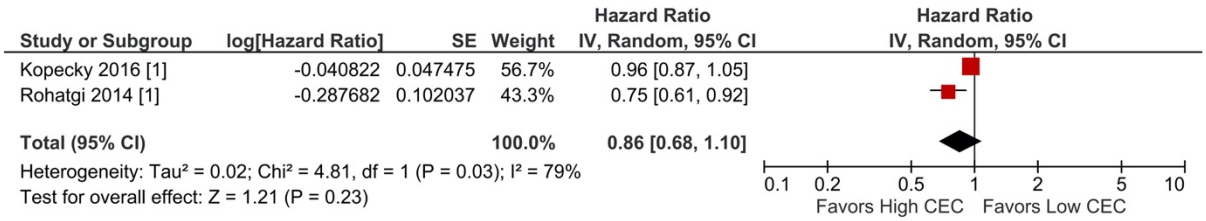

Figure S5. Atherosclerotic Cardiovascular Disease: Per SD Increment of CEC (Adjusted HR) (6 studies)

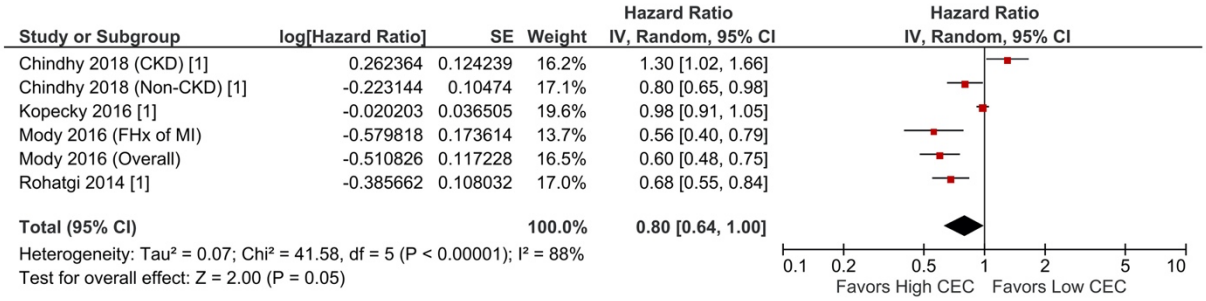

Figure S6. Atherosclerotic Cardiovascular Disease: Per SD Increment of CEC (OR) (5 studies)

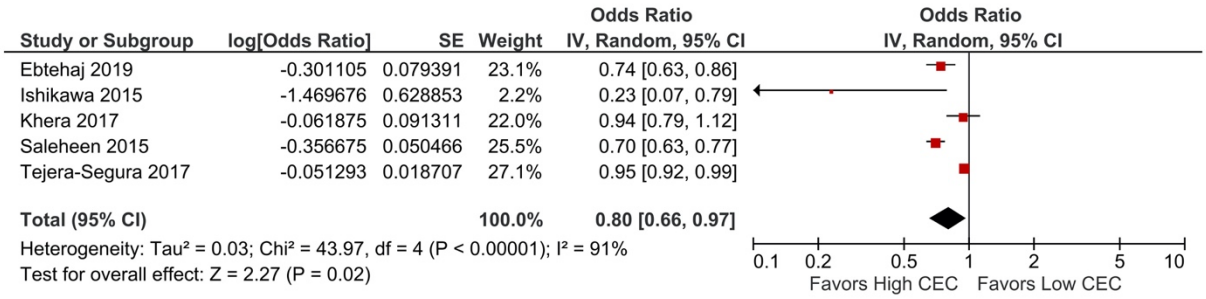

Figure S7. Atherosclerotic Cardiovascular Disease: Per SD Increment of CEC (Adjusted OR) (8 studies)

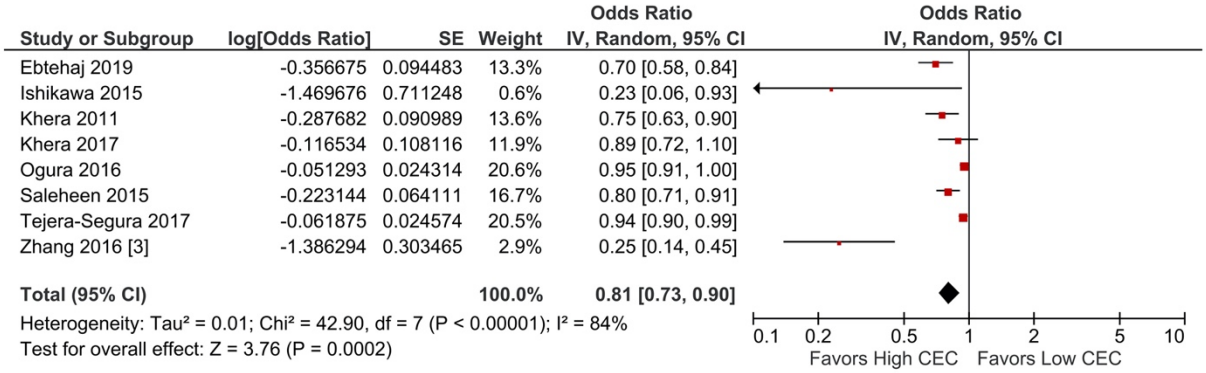

Figure S8. Death from All Causes: High CEC vs. Low CEC (RR) (3 studies)

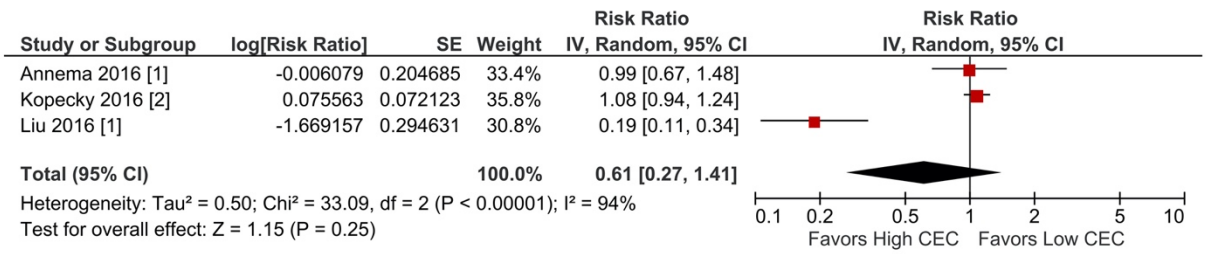

Figure S9. Death from All Causes: Per SD Increment of CEC (HR) (4 studies)

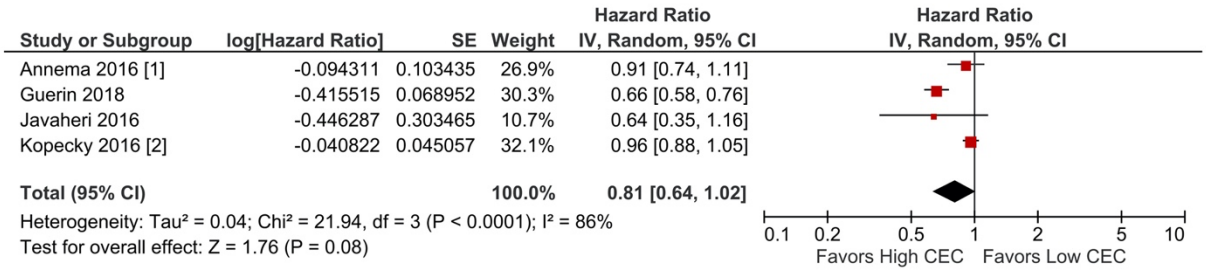

Figure S10. Death from All Causes: Per SD Increment of CEC (Adjusted HR) (5 studies)

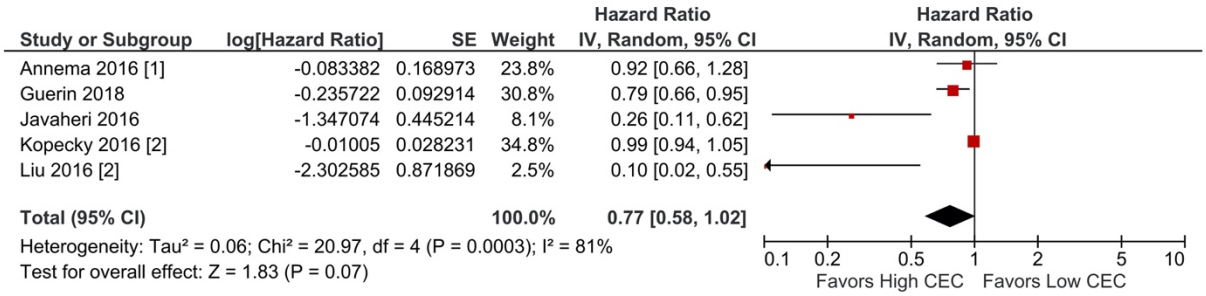

Figure S11. Death from Cardiovascular Causes: High CEC vs. Low CEC (RR) (4 studies)

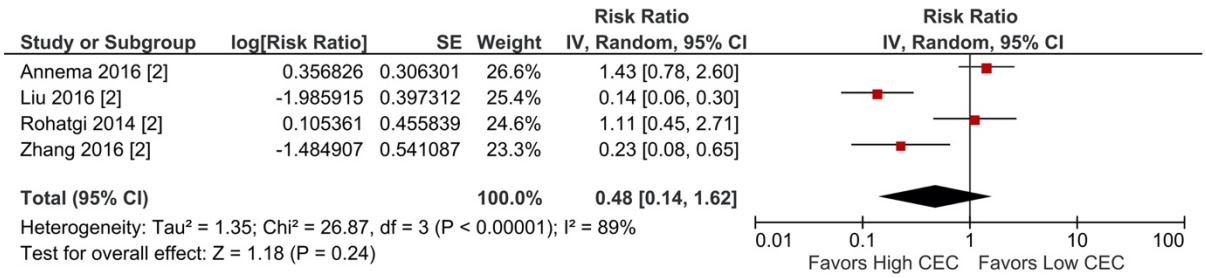

Figure S12. Death from Cardiovascular Causes: Per SD Increment of CEC (Adjusted HR) (4 studies)

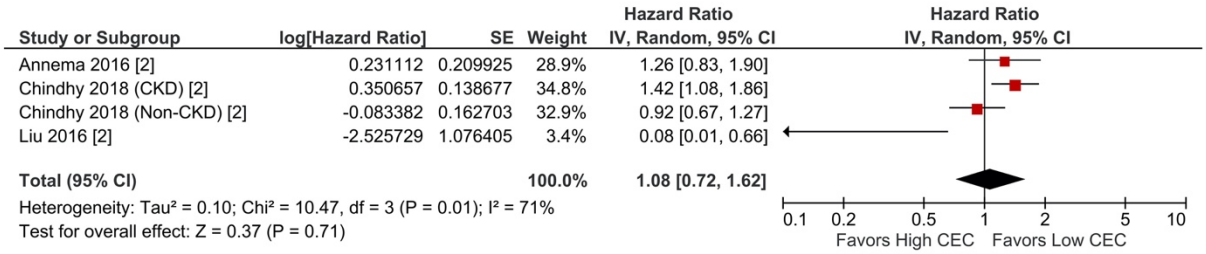

Figure S13. Adverse Cardiovascular Event: Mean Difference in CEC (6 studies)

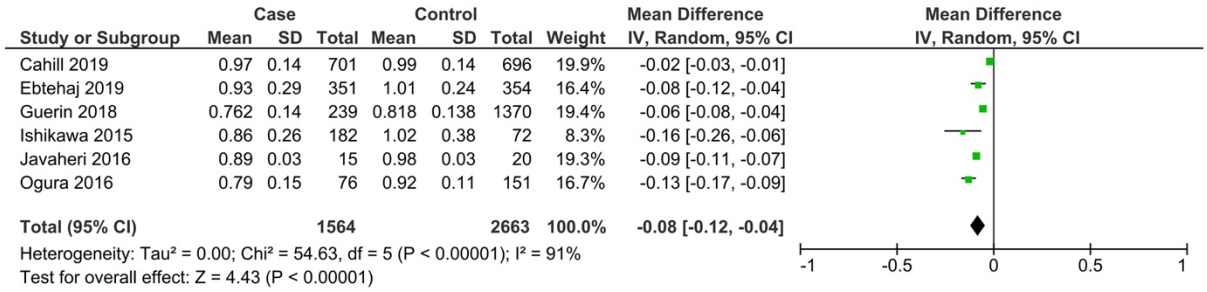

Figure S14. Atherosclerotic Cardiovascular Disease: Mean Difference in CEC (4 studies)

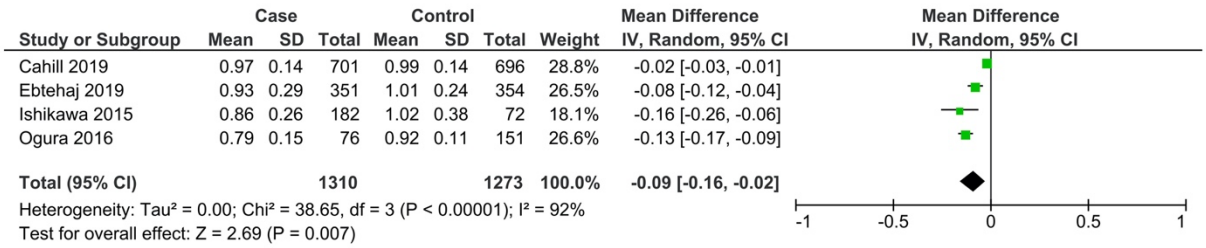

Figure S15. Death from All Causes: Mean Difference in CEC (2 studies)

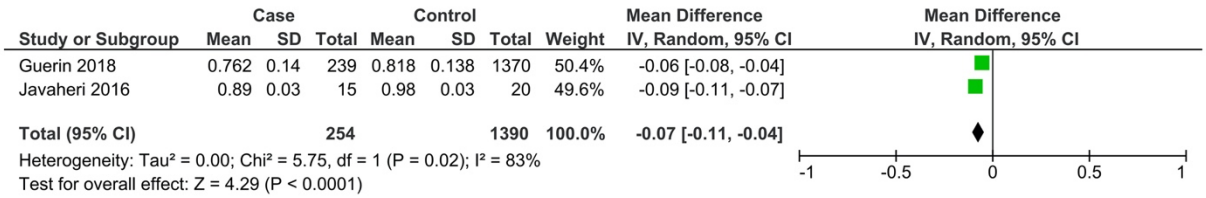

Figure S16. Funnel Plot of Adverse Cardiovascular Event: High CEC vs. Low CEC (RR) (14 studies)

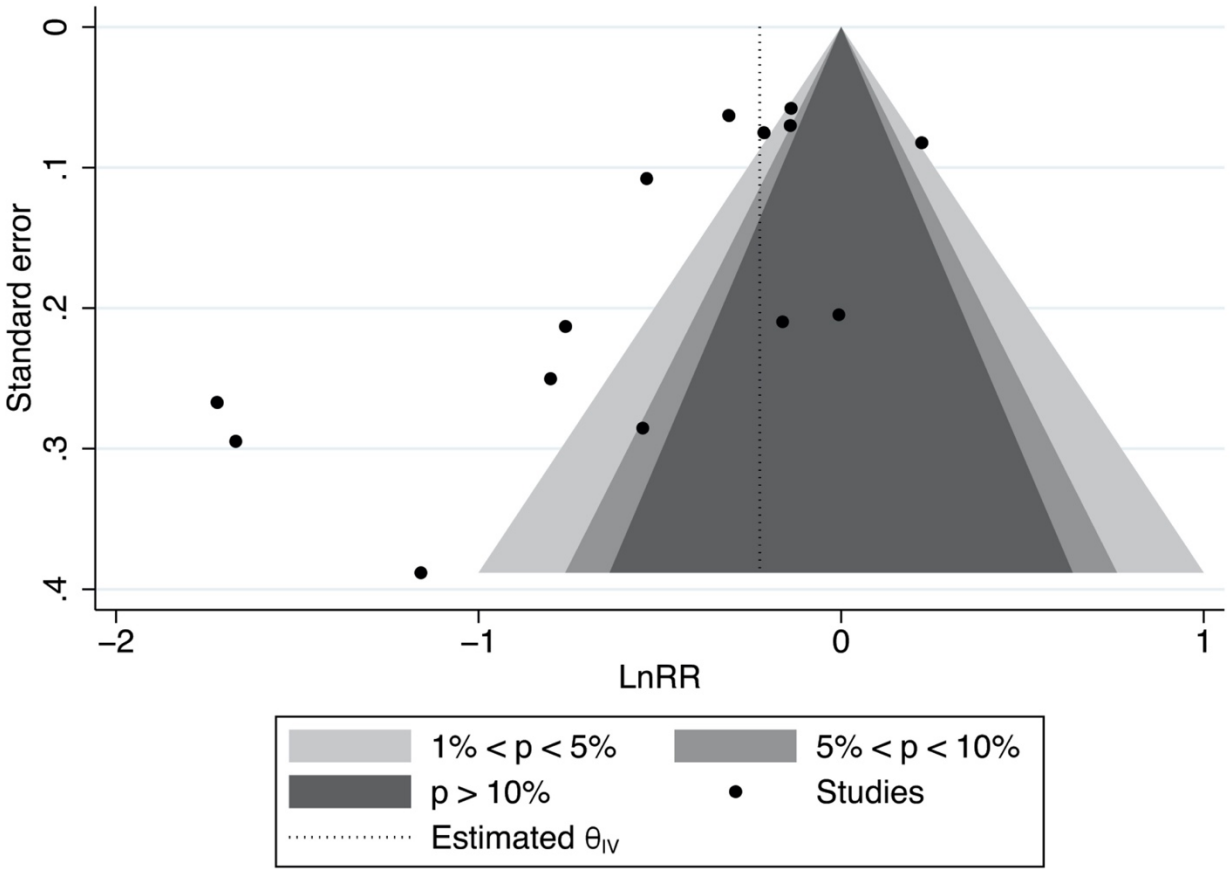

Figure S17. Funnel Plot of Adverse Cardiovascular Event: Per SD Increment of CEC (Adjusted HR) (10 studies)

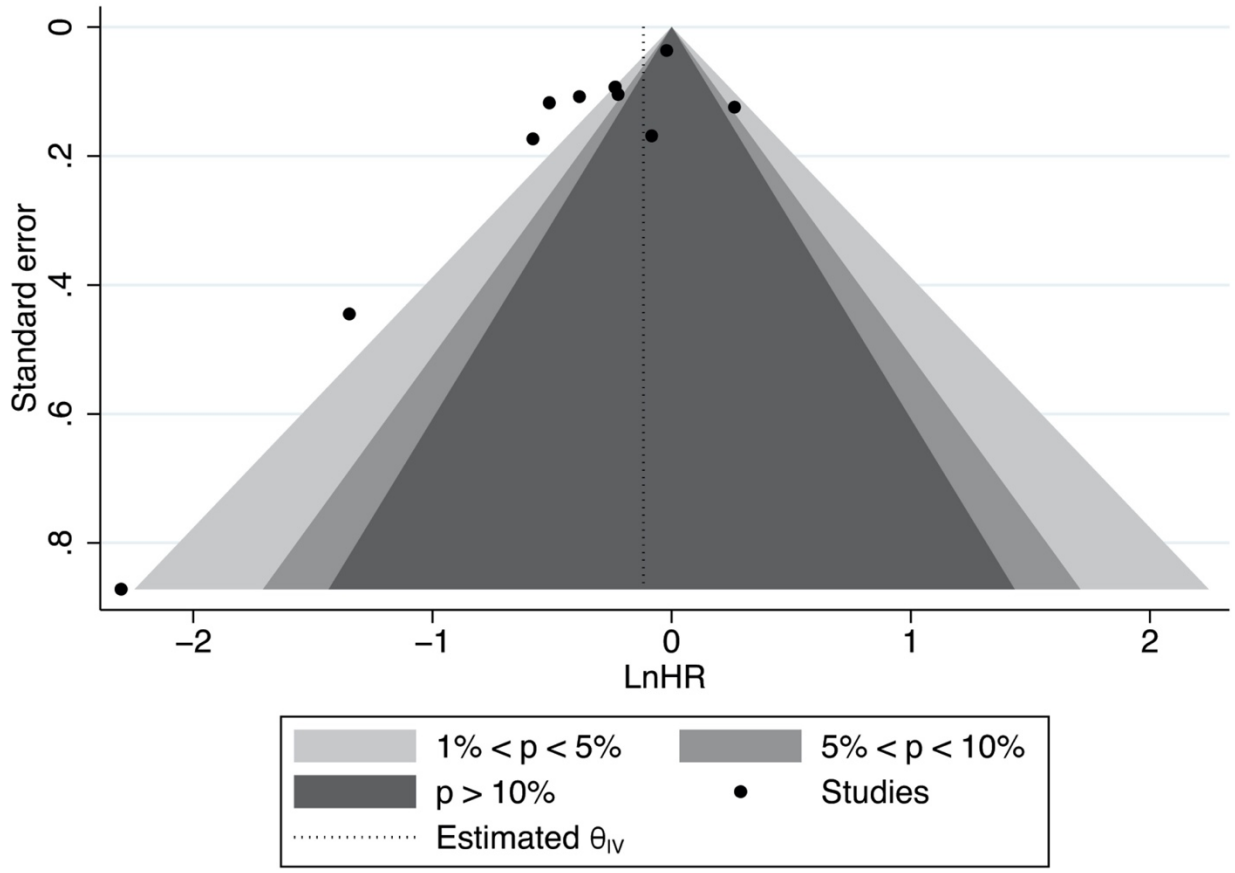

Figure S18. Funnel Plot of Atherosclerotic Cardiovascular Disease: High CEC vs. Low CEC (RR) (12 studies)

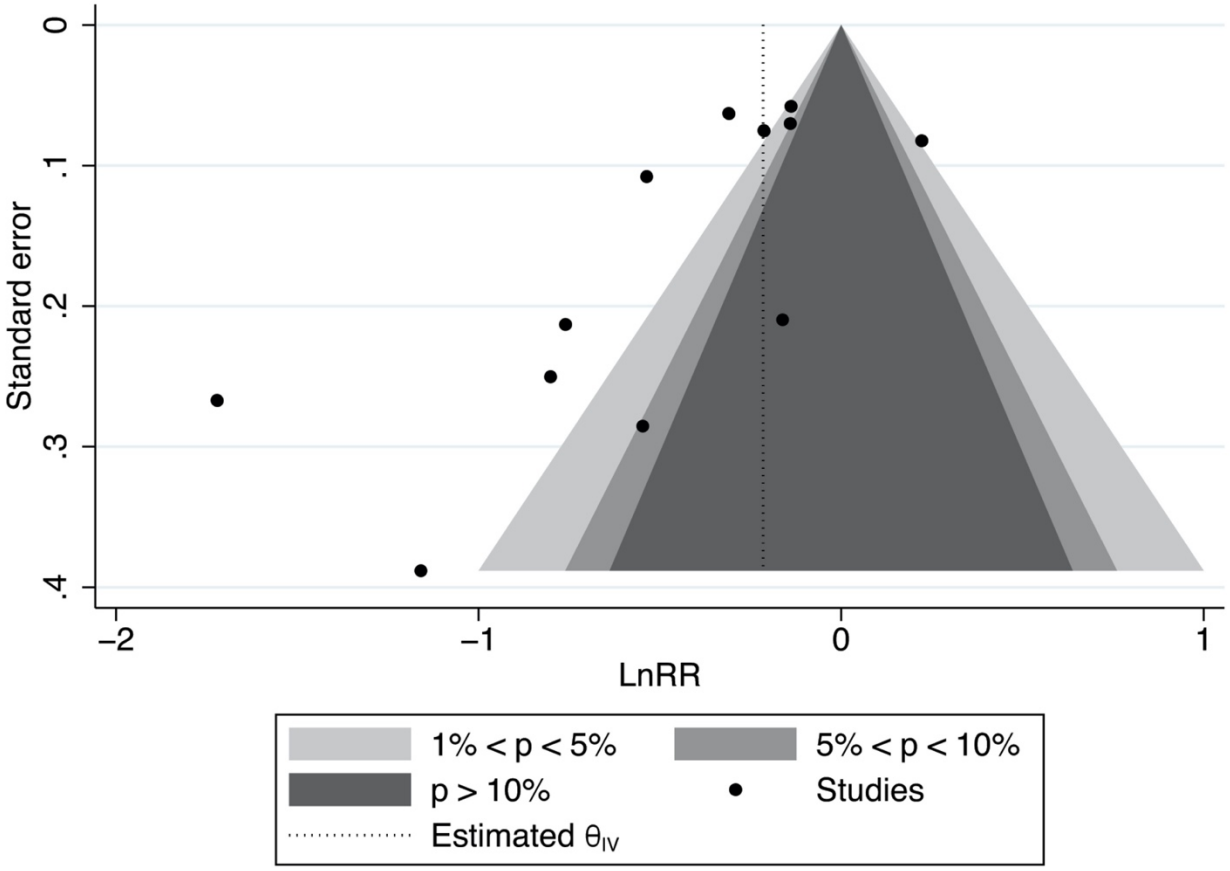

Figure S19. Subgroup Analysis of Adverse Cardiovascular Event: High CEC vs. Low CEC (RR) (14 studies)

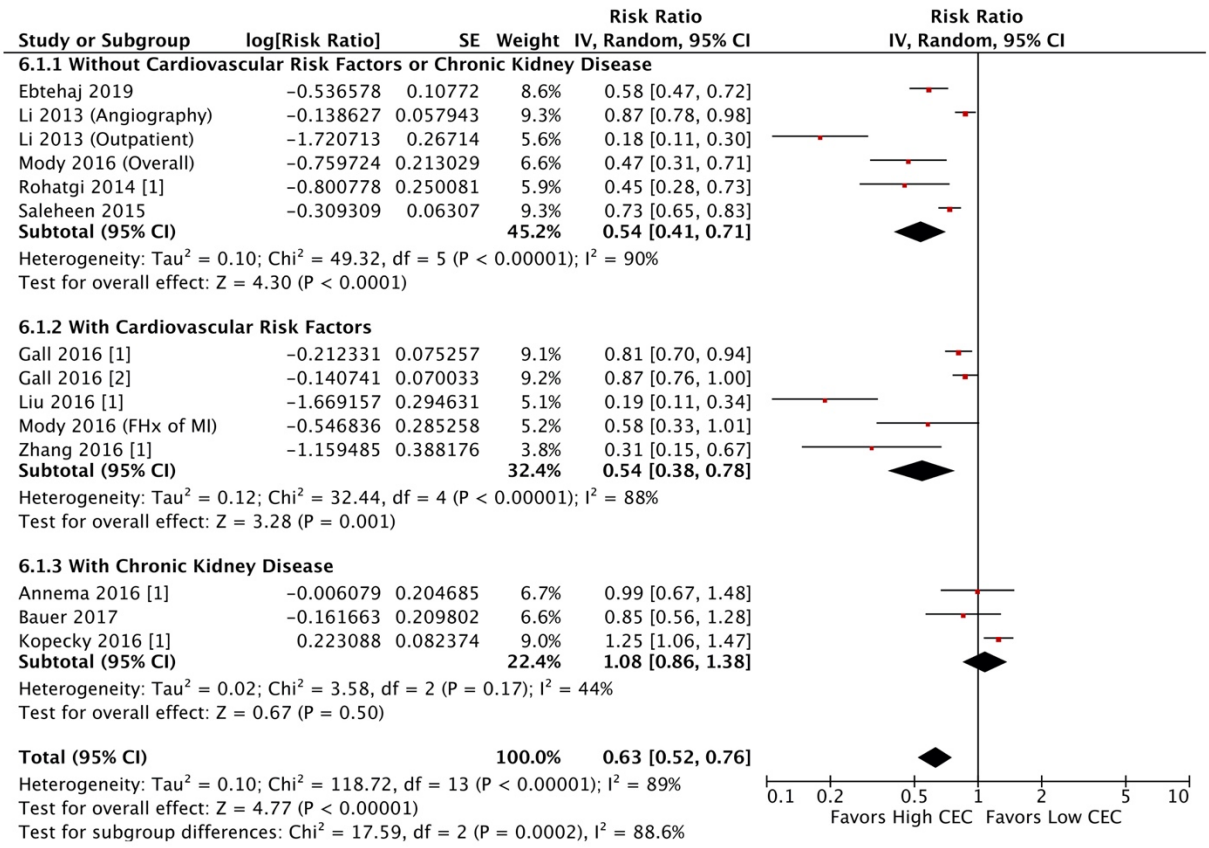

Figure S20. Subgroup Analysis of Adverse Cardiovascular Event: Per SD Increment of CEC (Adjusted HR) (10 studies)

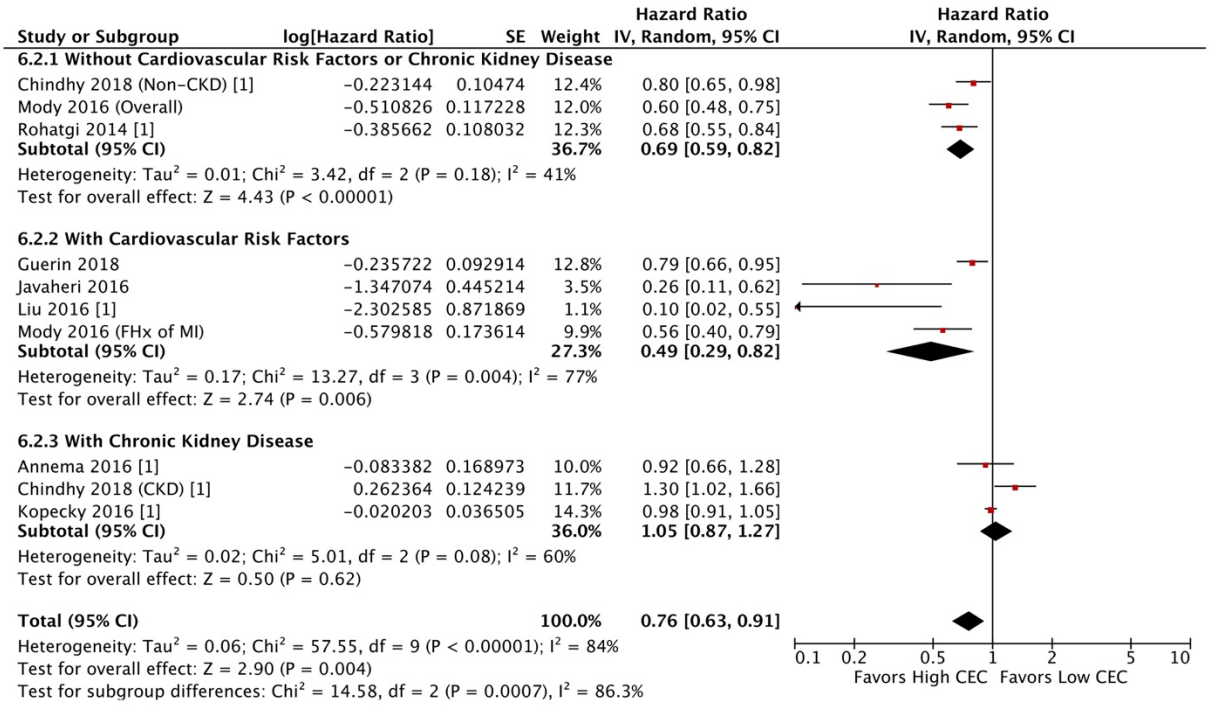

Figure S21. Subgroup Analysis of Atherosclerotic Cardiovascular Disease: High CEC vs. Low CEC (RR) (12 studies)

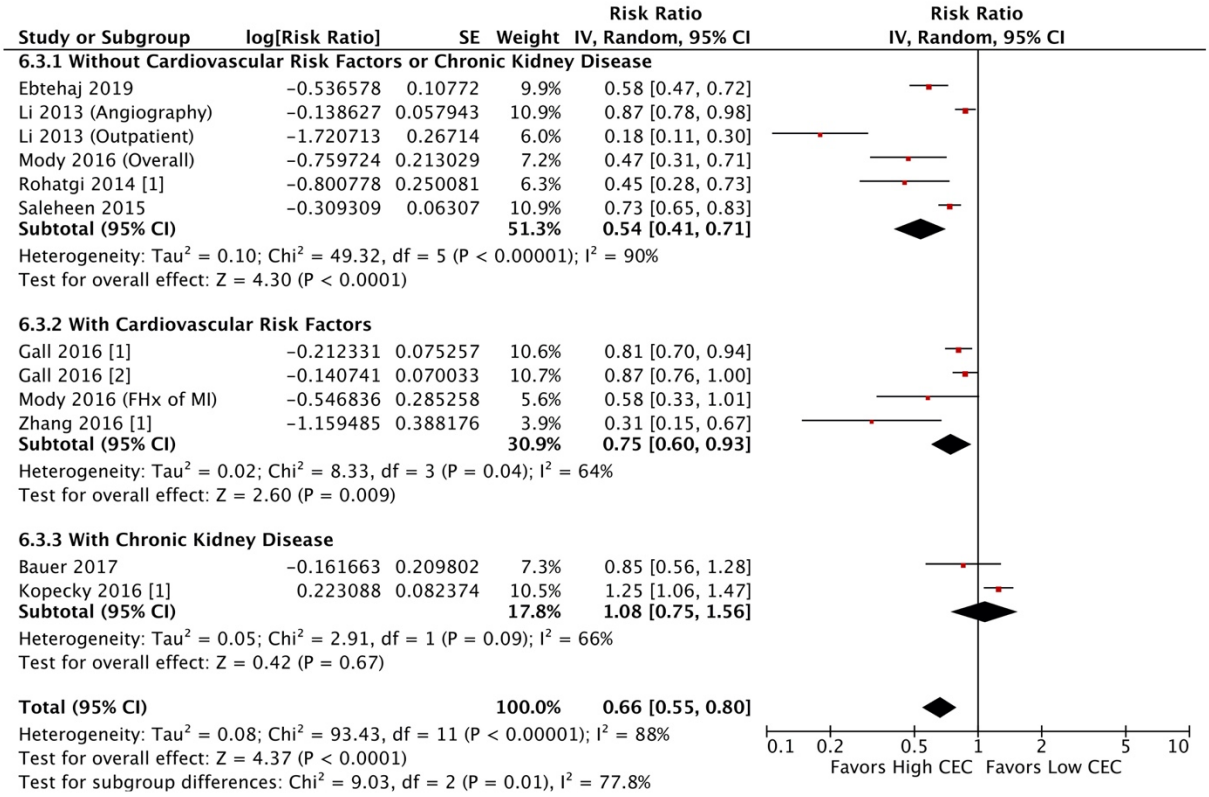

Figure S22. Adverse Cardiovascular Event Stratified by the Type of Cholesterol Donor: High CEC vs. Low CEC (RR) (14 studies)

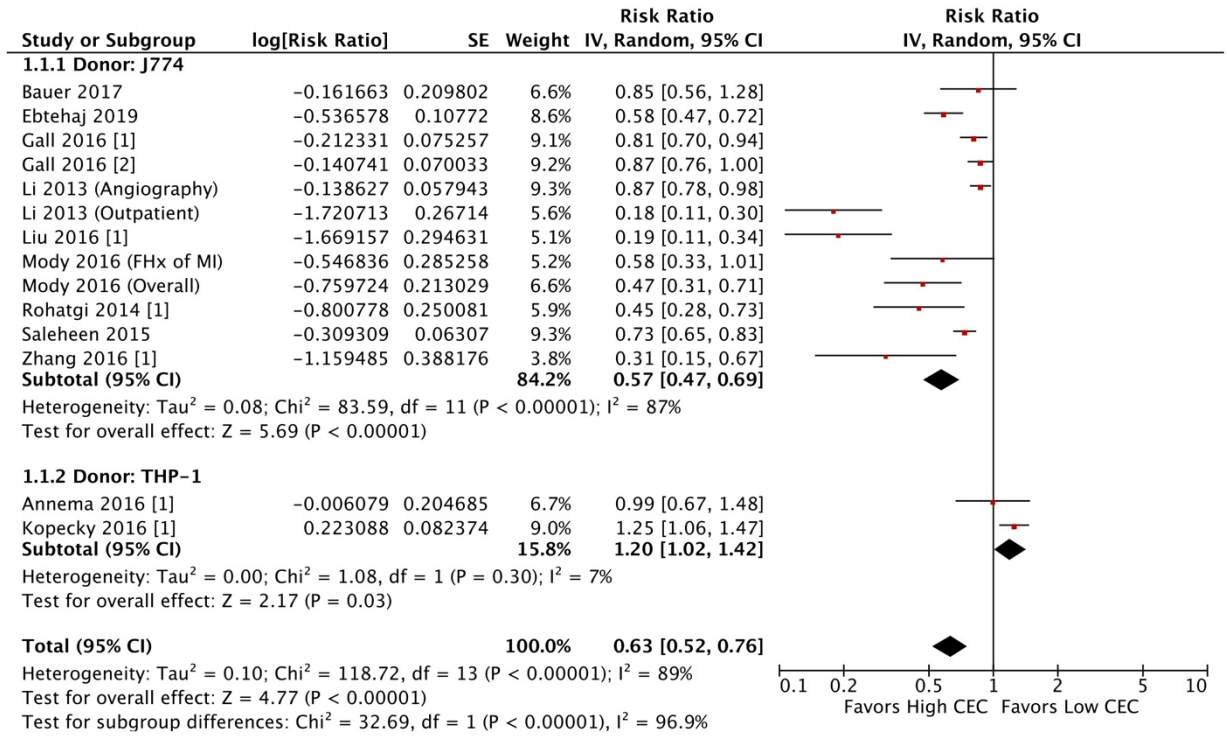

Figure S23. Adverse Cardiovascular Event Stratified by the Type of Cholesterol Donor: Per SD Increment of CEC (HR) (5 studies)

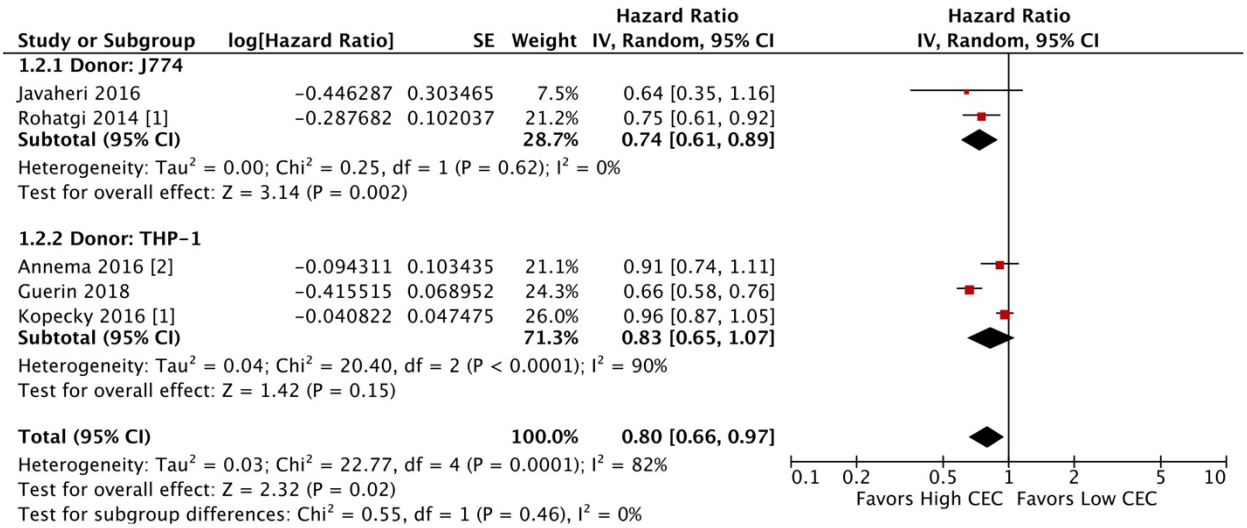

Figure S24. Adverse Cardiovascular Event Stratified by the Type of Cholesterol Donor: Per SD Increment of CEC (adjusted HR) (10 studies)

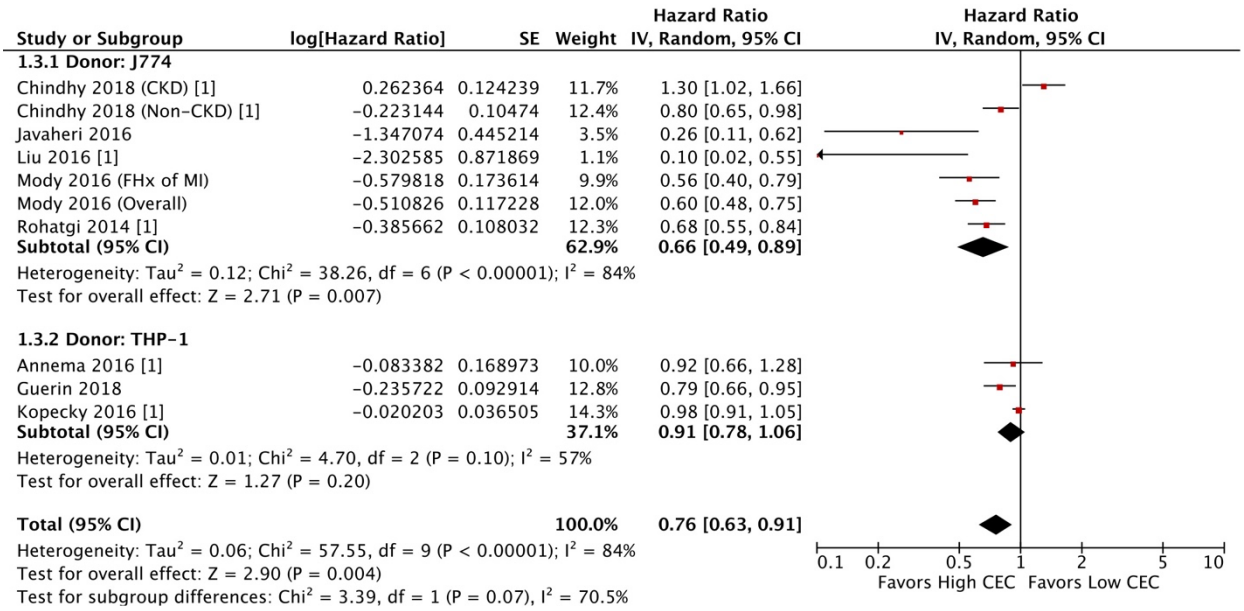

Figure S25. Adverse Cardiovascular Event Stratified by the Method of Labeling: High CEC vs. Low CEC (RR) (14 studies)

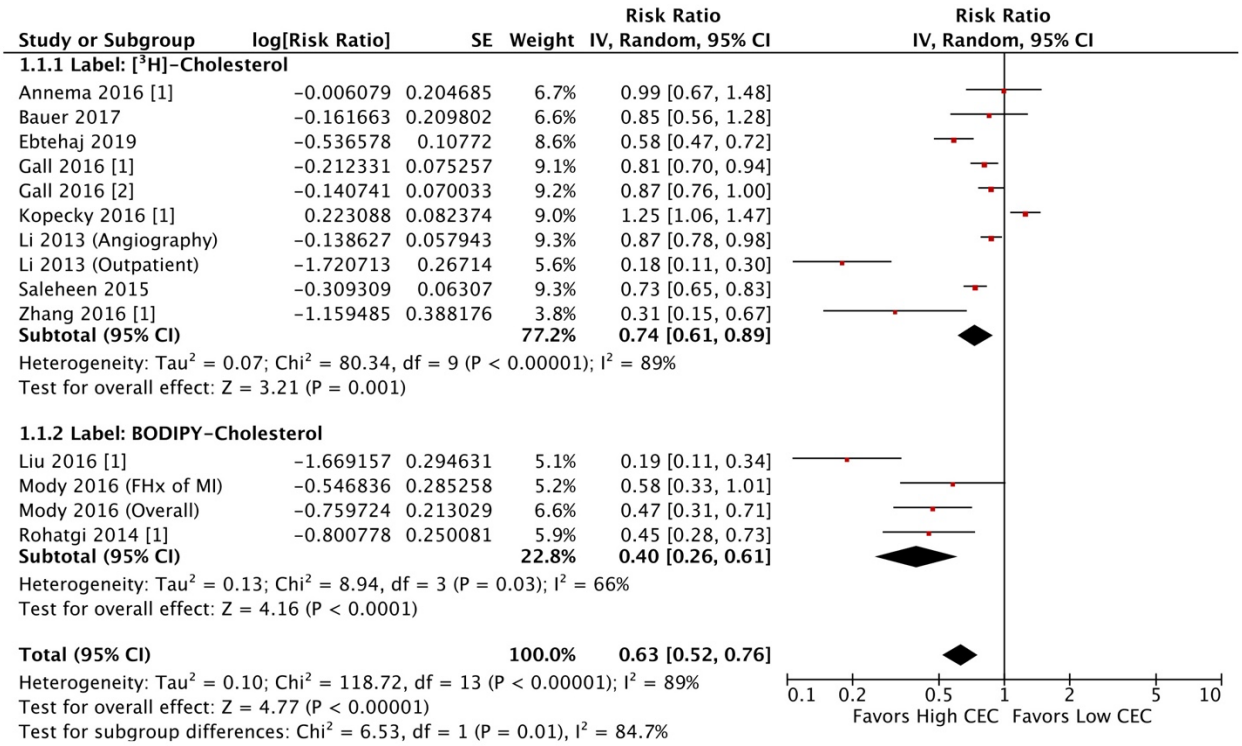

Figure S26. Adverse Cardiovascular Event Stratified by the Method of Labeling: Per SD Increment of CEC (HR) (5 studies)

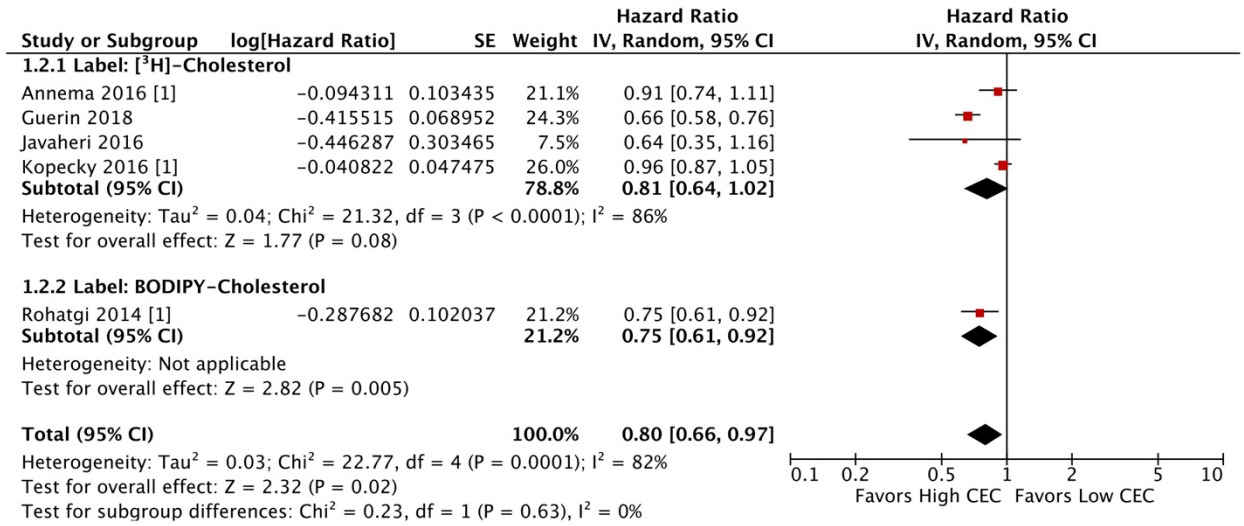
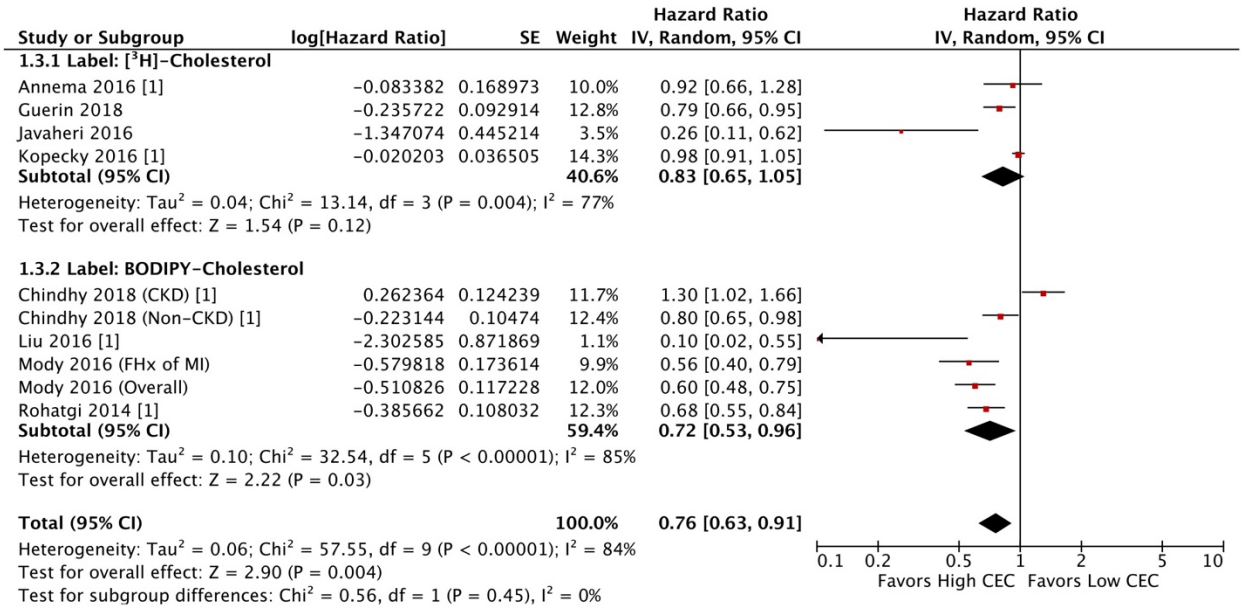
Figure S27. Adverse Cardiovascular Event Stratified by the Method of Labeling: Per SD Increment of CEC (adjusted HR) (10 studies)